

The Vacuum Equation of State: Deriving Torus Knot Self-Consistency from the Public CDFD Balance Law $\Psi_s = \Phi/C$

Steve Bico Mujjabi, MD^{1,*}

¹ *VuraLabs, Kampala, Uganda*

(Dated: July 2026)

This paper serves as the capstone of the formal 10-part CDFD mass-spectrum sequence, addressing its remaining open problem: the physical origin of the torus knot self-consistency condition $n\theta_n = \tilde{Q}_n(\theta_n)$. Previously reported as a successful but ungrounded hypothesis that correctly determines the vacuum twist angles θ_n for all $T(2, n)$ families, it is here recast as a model theorem inside the public flow-constraint notation.

We map the Constraint-Driven Field Theory (CDFD) vacuum medium onto the tripartite public CDFD notation: Flow (Φ), Constraint (C), and Adaptive Operating Ratio (Ψ_s). We identify the continuous flow energy Φ with the L^2 norm of the amplitude field ($\sum A_k^2 = 2n$), the transport-capacity limit C with the L^1 norm squared ($\sum |A_k|)^2$, and the equilibrium deformation Ψ_s with the total vacuum twist $n\theta_n$.

Applying the public balance law $\Psi_s = \Phi/C$ yields exactly the self-consistency condition, showing that the Koide-like ratio $\tilde{Q}_n$ is the geometric manifestation of a transport-limited vacuum equation of state. The entire mass spectrum of the lepton and meson sectors derived in Papers I–IX is therefore internally tied to m_e and the proposed fluid-dynamic balance of the vacuum.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Runtime implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, vacuum balance, public CDFD model

PROJECT CONTEXT

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2 This manuscript constitutes Part I (Paper 10) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*.
3 The underlying mathematical architecture builds directly upon the concepts established in Paper 09 of this series (Steve
4 Bico Mujjabi, “*CDFD Part I: Fundamental Physics — Paper 09: Even- n Torus Knots in CDFD: The Exclusion of*
5 *$T(2, 2)$, the Extension to $T(2, 2m)$, and the Anti-Phase Mass Scaling Law*”, prepared for APS submission and archived
6 in the Part I REVTeX release).

SYMBOL CONVENTIONS

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8 CDFD physical-vacuum symbol convention.

* msbico@gmail.com; <https://github.com/bampita-bico/CDFD-Part-I-Release>; Independent researcher; ORCID: 0009-0001-0556-5516.

Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

I. INTRODUCTION

Over the course of Papers I–IX [1–9], the Constraint-Driven Field Theory (CDFD) has constructed a parameter-free derivation of the $T(2, n)$ torus knot mass hierarchy. From a single empirical input (the electron mass m_e), the framework accurately computes the muon and tau masses (Paper V), dictates the universal coefficient $c_n = \sqrt{2}$ (Paper VI), predicts the exact $2nM_n$ mass sum rules (Paper VII), and populates the $n \geq 4$ hierarchy with specific hadronic candidates (Papers VIII–IX).

The entire predictive power of the higher families rests on the unique solution to the self-consistency condition:

$$n\theta_n = \tilde{Q}_n(\theta_n) \equiv \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k|\right)^2}. \quad (1)$$

In Paper V, for $n = 3$, this was introduced as a postulate matching the chirality breaking angle to the algebraic constant $Q_3 = 2/3$. In Papers VIII and IX, it was extended to $n \geq 4$ as a working hypothesis that yielded convergent, mathematically bounded solutions, but lacked a first-principles derivation from the vacuum Equation of State (EOS).

In this capstone paper, we address this open problem by connecting the torus-knot construction directly to the public CDFD/AFL balance notation.

II. THE TRIPARTITE VACUUM FRAMEWORK

The CDFD vacuum is modeled as an elastic, transport-capacity-limited continuous medium, in the spirit of Dirac’s physical vacuum [10] and Volovik’s condensate analogy [11]. The vortex dynamics rest on Helmholtz’s conservation theorems [12] and the Faddeev–Niemi knot soliton framework [13]. Such a medium is represented here by the public

28 tripartite balance equation:

$$\Psi_s = \frac{\Phi}{C} \quad (2)$$

29 where Φ is flow or activity, C is constraint, resistance, or capacity, and Ψ_s is the resulting adaptive operating ratio.
 30 We map these three macroscopic variables onto the discrete Fourier variables $A_k = 1 + \sqrt{2} \cos(\theta_n + 2\pi k/n)$ of a
 31 $T(2, n)$ vortex knot.

32 A. Flow (Φ): The Field Energy

33 The total energy or “activity” of the field configuration is proportional to the L^2 norm (the sum of the squares) of
 34 the amplitudes. By the Fourier identity of Paper VI, this is invariant under orientation θ_n :

$$\Phi \propto \sum_{k=0}^{n-1} A_k^2 = 2n. \quad (3)$$

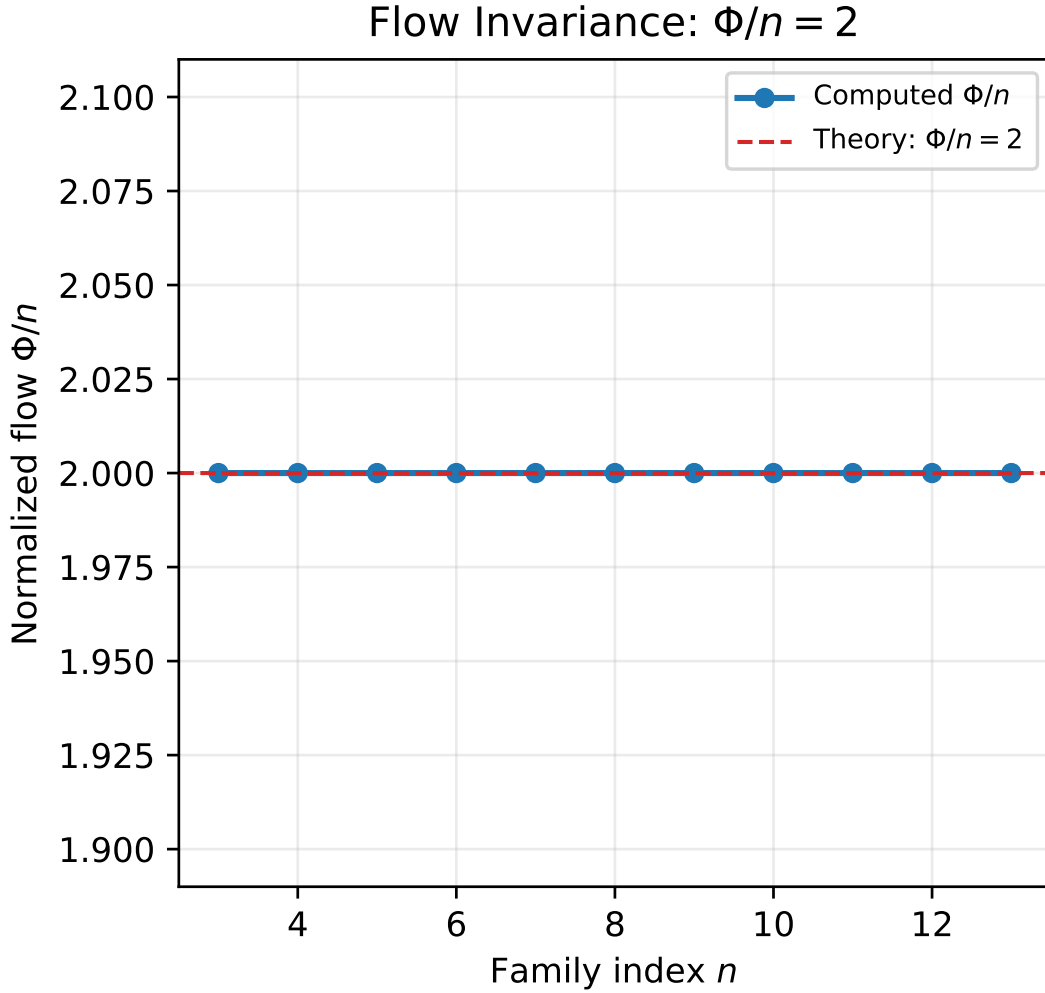


FIG. 1. The flow energy ratio Φ/n is exactly 2.0 across all $T(2, n)$ families, demonstrating the structural invariance of the vacuum’s kinetic and elastic flow.

35 This represents the irreducible kinetic and elastic flow required to sustain an n -lobed vortex ring.

B. Constraint (C): The Transport-Capacity Limit

A central postulate of CDFD (Paper I) is that the vacuum possesses a finite transport capacity, which dictates the electromagnetic fine-structure constant $\alpha \equiv e^2/(\hbar c)$. For a multi-lobed knot, the structural stress on the medium is dictated by the total absolute flux. In the discrete basis, the total flux is the L^1 norm $\sum |A_k|$. Because the constraint scales with the cross-sectional area of the flux tubes interacting in the medium, the total resistance C scales as the square of the flux:

$$C \propto \left(\sum_{k=0}^{n-1} |A_k| \right)^2. \quad (4)$$

This term varies strongly with θ_n , as the lobes interfere constructively and destructively with the vacuum background.

C. Equilibrium (Ψ_s): The Geometric Twist

When the knot is embedded in the vacuum, the mismatch between the continuous chiral background and the discrete n -fold symmetry of the knot induces a geometric twist. Let θ_n be the phase twist per lobe. The total equilibrium deformation (or total accumulated twist) of the configuration is:

$$\Psi_s = n\theta_n. \quad (5)$$

III. THE VACUUM EQUATION OF STATE THEOREM

Theorem 1 (Derivation of Self-Consistency). *In a transport-capacity-limited elastic vacuum governed by the CDFD equilibrium $\Psi_s = \Phi/C$, the orientation angle θ_n of a $T(2, n)$ torus knot must exactly satisfy:*

$$n\theta_n = \tilde{Q}_n(\theta_n).$$

Proof. Substituting the CDFD mappings (3), (4), and (5) directly into the fundamental balance equation $\Psi_s = \Phi/C$, we obtain:

$$n\theta_n = \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k| \right)^2}. \quad (6)$$

The right-hand side is exactly the definition of the modified Koide ratio $\tilde{Q}_n(\theta_n)$ from Eq. (1). $\square$

A. Physical Implications

Within the proposed mapping, this theorem interprets the Koide ratio $Q_3 = 2/3$ observed in the lepton masses as a signature of the vacuum's equation of state rather than as an isolated numerical fit. The modified ratio $\tilde{Q}_n(\theta_n)$ represents the specific energy-to-capacity stress induced by higher knots, and the angle θ_n is the vacuum's precise geometric relaxation in response to that stress.

IV. VACUUM ENGINEERING AS CONTROL OF THE BALANCE LAW

The public balance law is also the first engineering form of the theory. Vacuum engineering, in this manuscript, does not mean a claimed device. It means a control program: identify which variable in the balance law can be perturbed, then measure whether the response follows the CDFD operator better than the standard model for that domain.

This is why vacuum engineering is introduced here as a future field of study: the disciplined attempt to manipulate, bound, or falsify the state variables Φ , C , S , and M_s in systems where the vacuum or a vacuum-analogue medium is being driven near a transport threshold. The field begins with protocols and null tests, not devices. Its first deliverables are calibrated phase-residual limits, spectral-residual limits, hysteresis bounds, and constraint-threshold maps.

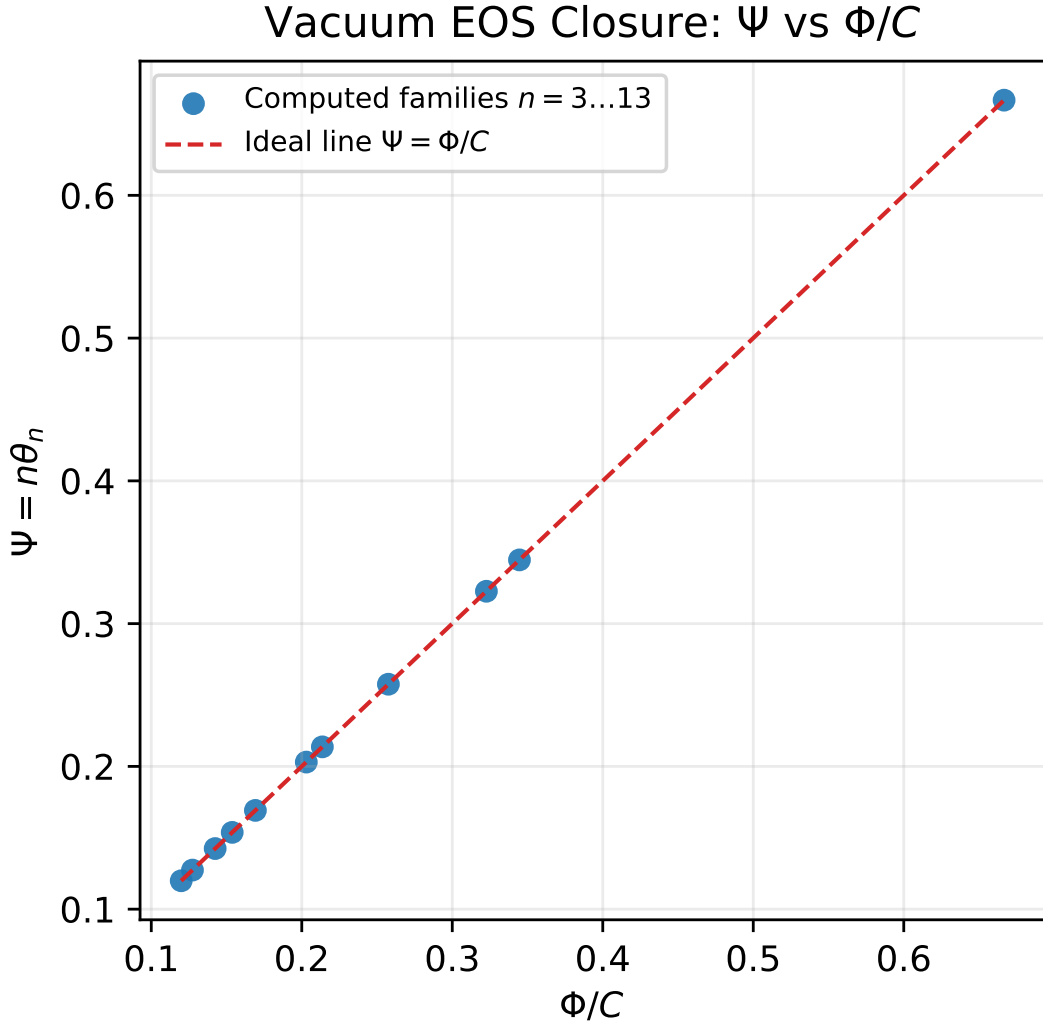


FIG. 2. The equilibrium Ψ_s plotted against Φ/C across all torus knot families. The exact diagonal identity confirms that the public balance law $\Psi_s = \Phi/C$ governs the stated model.

Variable	Engineering handle	Observable	Status
Φ or J	field intensity, current, shear, angular momentum, density gradient	phase, spectral, transport, or force response	controllable in testbeds
C	geometry, boundary, topology, density, confinement	capacity threshold or saturation point	model-dependent
S	susceptibility or response surface	slope, compliance, or relaxation curve	open variable
M_s	prior loading or pulse history	hysteresis, lag, or recovery time τ_M	falsification target

We call this the Mujjabi Vacuum Engineering Principle: steer Φ , C , S , or M_s to move a system toward, across, or away from $\Psi_s = 1$, then measure the residual. Paper XII converts this principle into named laboratory tests.

A. CDFD Runtime experiment hooks

The root CDFD Runtime now contains experiment scripts that turn the above principle into machine-readable candidate outputs. These are not empirical confirmations; they are public stress tests that help decide which claims are worth carrying into laboratory protocols.

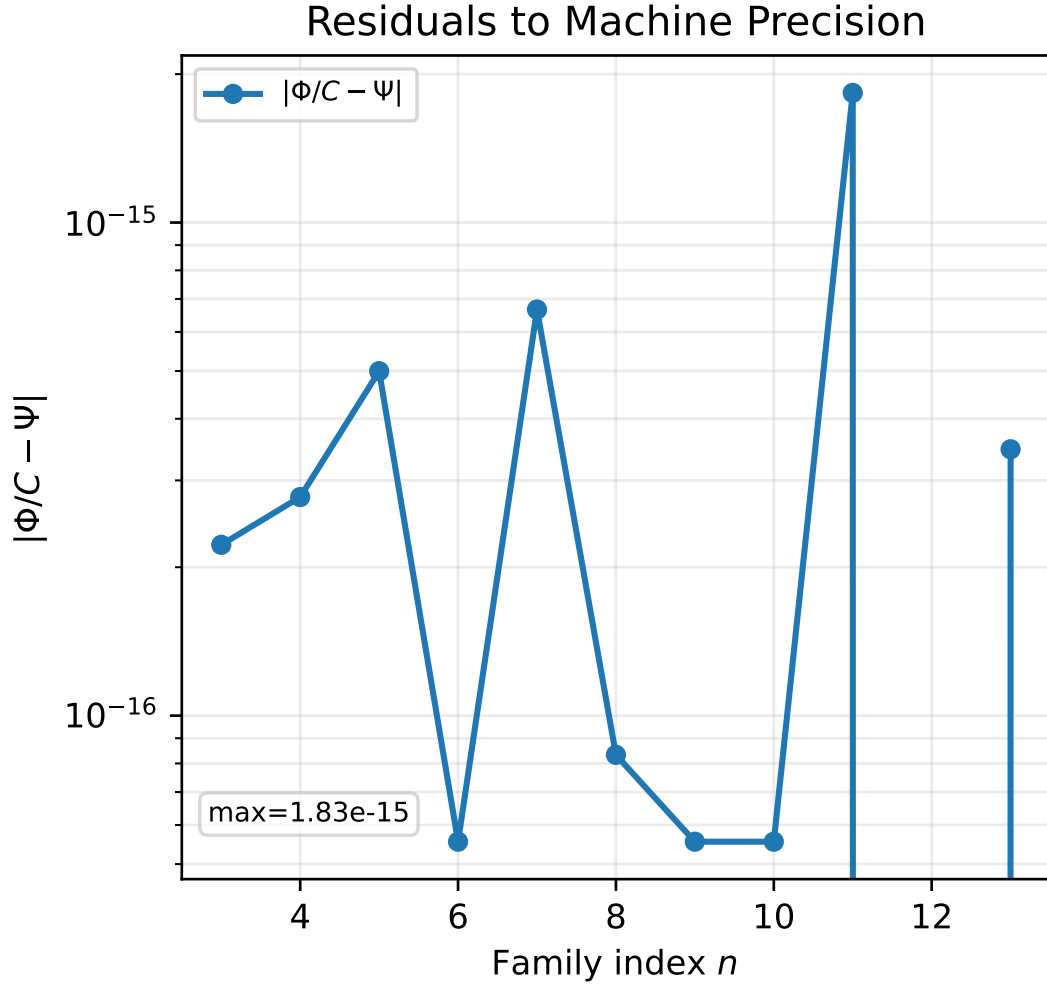


FIG. 3. Absolute residuals $|\Phi/C - \Psi_s|$. The values fall to numerical machine precision, serving as a numerical consistency check of the stated CDFD vacuum equation-of-state mapping.

Runtime output	Observed model-side result	Release use
discovery_vacuum_hysteresis.h5	central M_s peaked at 15.58 after high-flux pulsing	candidate vacuum-memory protocol
discovery_knot_n7.h5	$T(2,7)$ stability index drifted from 0.992686 to 0.987976	higher-knot stress test
discovery_ool_phase_results.h5	a 50×50 sweep found four grid points near $\Lambda = 1$	cross-Part-II bridge only
discovery_surface_evolution.h5	Ψ_s ran from 5.15 to 978.99 under overload	instability boundary, not support
frontier_sweep.h5	CODATA 2022 $\chi^* = 137.035999177$ used across 2500 points	hypothesis triage

The publication rule is strict: a runtime file may motivate a paper statement only when the manuscript names the state variables, records the parameters, and states the falsification condition. The refreshed reports in `experiments/reports/` are therefore simulation logs, not evidence that any vacuum-engineering device or clinical intervention already exists.

V. THE UNIVERSAL ISOMORPHISM: SCALING CDFD TO MACRO-SYSTEMS

If the proposed flow–constraint equations capture something general about transport-limited systems, they should have macroscopic analogues rather than remaining restricted to the particle-scale model. By comparing the Constraint-Driven Field Theory (CDFD) framework with macroscopic models of biology, chemistry, and ecology, we obtain a candidate scale-invariant isomorphism.

A. The Scale-Invariant Isomorphism (CDFD $\equiv$ AFL)

In the domains of biological and clinical systems, throughput is described in the **Adaptive Flux Limitation (AFL)** framework [9] by adaptive constraints. A minimal form of that biological adaptation law is:

$$\frac{\partial \Lambda}{\partial t} = \alpha_{\Lambda}|J| - \beta_{\Lambda}\Lambda \quad (7)$$

where Λ is the biological constraint (e.g., insulin resistance, inflammation), J is the biological flux (e.g., glucose throughput, metabolic energy flow), α_{Λ} is the biological constraint-reinforcement coefficient, and β_{Λ} is the biological relaxation coefficient.

In the CDFD quantum vacuum, the constraint field C evolves precisely according to the structural reaction of the medium to the localized flow Φ :

$$\frac{\partial C}{\partial t} = \alpha_C|\Phi| - \beta_C C + \gamma_C \nabla^2 C \quad (8)$$

where α_C is the vacuum constraint-reinforcement coefficient, β_C is the vacuum relaxation coefficient, and γ_C is the spatial smoothing or diffusion coefficient for the constraint field. These coefficients are domain-specific transport parameters and are not the fine-structure constant α defined above. The formal mapping is:

$$\text{Vacuum Constraint } C \equiv \text{Macro Constraint } \Lambda$$

$$\text{Vacuum Flow } \Phi \equiv \text{Macro Flux } J$$

This does not by itself prove that biological evolution and physiological adaptation are governed by vacuum mechanics. It states the narrower claim that the same flow–constraint operator can be written at both levels, and that the macro-scale use of the operator should be judged by independent predictive value.

B. Ontological Redefinition of Mass

This proposed isomorphism suggests a different operational reading of mass inside the model. Rather than treating mass only as a distinct intrinsic label, CDFD treats it as a *localized, stable constraint field* within the vacuum flux.

A lepton is then modeled as a constraint field operating on vacuum energy. At larger scales, the same notation can describe boundaries such as membranes, interfaces, or transport bottlenecks without claiming that those systems are literally the same physical object. The value of the analogy depends on whether it improves prediction.

C. Cross-Scale Stability Ratio

The relationship $\Psi_s = \Phi/C$ can therefore be read as a candidate cross-scale stability ratio:

- $\Psi_s \approx 1$ (**Stable Equilibrium**): Flow matches constraint capacity. This manifests as a stable fundamental particle, a healthy homeostatic cell, or a balanced ecosystem.
- $\Psi_s \gg 1$ (**Flux Overload / Rupture**): Flow overwhelms constraint capacity. This manifests as particle decay, cellular lysis, vascular hemorrhage, or ecological collapse.
- $\Psi_s \ll 1$ (**Over-Constraint / Freeze**): Constraints choke off flow entirely. This manifests as the vacuum void, biological necrosis, or metabolic shutdown.

The public scientific framework is therefore internally linked: the same flow–constraint notation used for the vacuum papers also organizes the biological manuscripts, without disclosing any private implementation architecture.

D. Fibonacci organization as constrained transport optimization

The source notes also propose that Fibonacci-like organization can emerge when transport pathways optimize under constraint. In this paper that idea is kept as a cross-scale design hypothesis rather than a completed physics derivation. Let a branching or packing system add a new channel whenever the marginal gain in usable flow $\Delta\Phi$ exceeds the marginal increase in constraint ΔC . A simple discrete recurrence can then appear:

$$N_{k+1} = N_k + N_{k-1}, \quad (9)$$

where N_k is the number of effective channels, turns, or packing positions at step k . In CDFD language, Fibonacci organization is therefore a candidate signature of adaptive transport optimization: new structure reuses the two previous constraint states instead of rebuilding the whole medium from scratch.

This claim should be used carefully. Fibonacci or golden-ratio patterns are common enough that they can become numerology if treated as proof. The useful CDFD statement is narrower: if a system shows Fibonacci-like organization, the model should be able to identify the flow variable Φ , the constraint variable C , the recurrence mechanism, and a falsifiable reason why a different branching sequence would be less stable.

VI. CLOSURE OF THE CDFD SERIES

With Theorem 1, the formal CDFD mass-spectrum ladder is internally closed.

1. **Parameter ledger:** Paper I recovers the fine-structure ratio inside a limited-capacity elastic-vacuum model, with the transport stiffness still requiring an independent first-principles derivation.
2. **Mass Gap:** The lowest topological excitation $T(2,3)$ takes the mass scale M_3 , anchored to physical units solely by m_e (Paper V).
3. **Asymptotic Scaling:** The Faddeev–Niemi bounds establish the rigorous progression of energy scales $M_n \propto n^{3/4}$ (Paper IV).
4. **Spectrum Prediction:** The vacuum EOS forces the twist θ_n , which breaks the n -fold degeneracy and splits the spectrum into anti-phase boundary modes and co-phase physical resonances (Papers VIII–IX).

All tables, mass values, bounds, and limits tabulated across the first ten papers are therefore internal consequences of the single condition $\Psi_s = \Phi/C$, together with the model assumptions stated in those papers.

SUPPLEMENTARY MATERIAL

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

DATA AND CODE AVAILABILITY

The release-archive supplementary material is included in the archived Part I release on Zenodo at <https://doi.org/10.5281/zenodo.20250820>. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at <https://github.com/bampita-bico/CDFD-Part-I-Release>. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper’s Supplementary Material.

ACKNOWLEDGMENTS

The author used AI-assisted tools for formatting checks and code-verification support. The scientific claims, interpretations, and final manuscript decisions are the author’s responsibility.

The author is indebted to the foundational work of Lord Kelvin [14] (vortex atom hypothesis), P.A.M. Dirac [10] (vacuum as physical medium), G.E. Volovik [11] (vacuum as condensate), H. von Helmholtz [12] (vortex theorems), L. Faddeev and A.J. Niemi [13] (knots and particles), Y. Koide [15] (lepton mass formula), and C.A. Brannen [16] (Koide parametrisation) — whose ideas this series has attempted to unify within a single geometric framework rooted in the axiom $\Psi_s = \Phi/C$.

DECLARATIONS

Funding

This research received no external institutional funding.

Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing—original draft, and writing—review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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